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The Complete Low-ESL Capacitor Guide



Pulse

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[Resistors](#), capacitors, and inductors... they're fundamental components and your electronics classes always imply that these components function exactly as described in textbooks. Unfortunately, that simply isn't true; your capacitor will eventually behave like an inductor at high frequencies, leading to unwanted behavior and incorrect impedance in your circuits.

The culprit is equivalent series inductance or ESL. All [capacitors](#) have some parasitic ESL that becomes measurable at high enough frequencies, and it's only a question of whether the ESL value matters for your specific application. High-speed digital systems, RF systems, and many other applications specifically

require low-ESL capacitors to set target impedance, filter within the desired frequency range and ensure decoupling in a PCB's PDN.

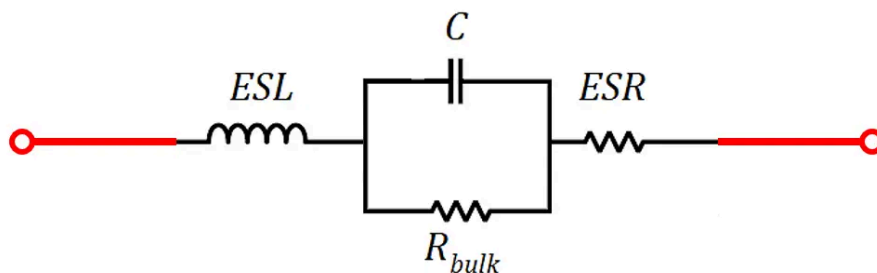
Some component datasheets or application notes simply state that you must use a certain type of capacitor without any greater explanation, while other datasheets will ask for a capacitor with a specific ESL value but without any other guidance. So how can you be sure you're using the right low-ESL capacitor in your design? The guidelines I've compiled here should help you get started understanding how to find and select low-ESL capacitors for advanced applications.

What is Low ESL?

All [components](#) have some parasitics, meaning some unintended inductance, resistance, and capacitance. These parasitics cause the real electrical behavior of a component to be different from the ideal behavior of the component. They can arise due to the construction of the component itself, or due to the way the component is placed on a PCB. In general, when supplied with DC power, passives will behave as ideal components, but parasitics start to take over electrical behavior at high frequencies.

Capacitor Circuit Model with ESL

In a capacitor, equivalent series inductance (ESL) is the apparent inductance in a capacitor, which only becomes noticeable beyond certain frequencies. There is also some equivalent series resistance (ESR). Finally, there is some leakage or bulk resistance in the capacitor, which exists in parallel with the ideal capacitance, ESL, and ESR. This is shown in the following image, as well as the true capacitor impedance.



$$\text{Impedance: } Z = \left(\frac{i\omega C}{1 - \omega^2 C \cdot \text{ESL} + i\omega C \cdot \text{ESR}} + \frac{1}{R_{\text{bulk}}} \right)^{-1} \approx \frac{1 - \omega^2 C \cdot \text{ESL} + i\omega C \cdot \text{ESR}}{i\omega C}$$

$$\text{Self-resonance: } \omega_r = \frac{1}{\sqrt{C \cdot \text{ESL}}}$$

Because the dielectric material in the capacitor is strongly insulating, the value of R_{bulk} is normally very large (~100 GOhms), so it can be ignored when calculating the capacitor's impedance. Therefore, we need to focus on the ESL and ESR values when [selecting capacitors](#).

Self-Resonance and ESL

If you look at the above circuit model, you'll see that a real capacitor is an RLC circuit, so it has some [self-resonant frequency](#) as defined above. Similar RLC models are used to describe the real behavior of [inductors](#), [transformers](#), and even [semiconductors](#) like diodes and transistors. This self-resonant frequency is the reason why real capacitors can act like inductors; when the driving frequency is larger than the self-resonant frequency, the inductive behavior of the component dominates.

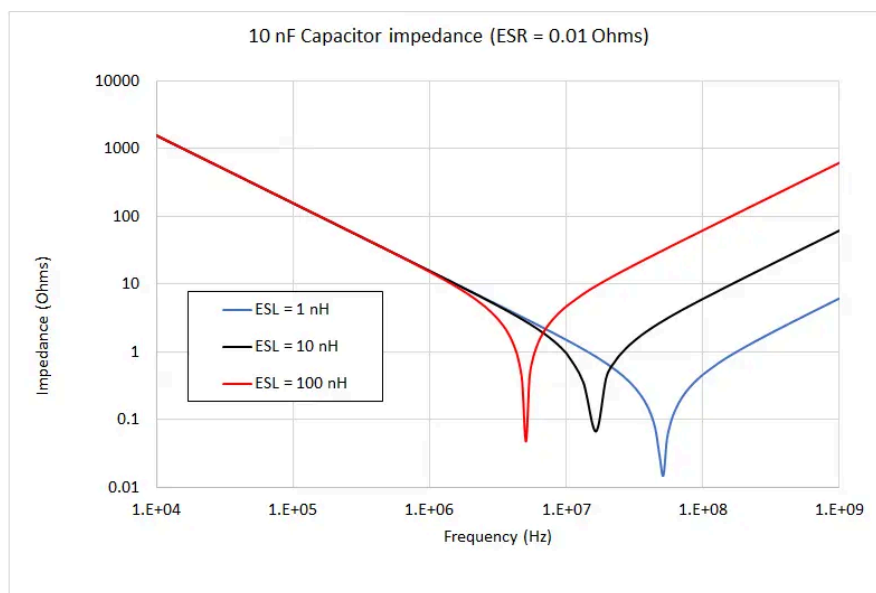
Why Have Low ESL and ESR?

In general, you can never have a capacitor with zero ESL and ESR, but some applications demand very low values.

There are three reasons you want low ESL values when selecting a capacitor, particularly for high speed/high-frequency applications:

- **In filtering applications:** Low ESL means the self-resonant frequency is higher, so the capacitor behaves like an ideal component over broader frequencies.
- **In power applications:** the transient response will be faster, meaning the capacitor can discharge and deliver power faster. The same benefits for [filtering also apply in power applications](#). Low ESR is also important here as charging/discharging is faster when ESR is lower.
- **In decoupling applications:** When used for decoupling/bypassing on high-speed ICs, low-ESL capacitors provide a greater reduction in ground bounce and supply bounce.

The image below shows how ESL affects the impedance of a theoretical 10 nF capacitor with 0.01 Ohms ESR. The various curves show impedance profiles for different ESL values (1 nH, 10 nH, and 100 nH). From the graph, we see that the impedance is capacitive up to the self-resonant frequency, regardless of the ESL value and then becomes inductive beyond the self-resonant frequency. We see that the impedance



For capacitors used in applications like switching [power supplies](#), inverters, or power converters, ESL is generally not such a major problem. PWM driver signals are generally slow enough that the vast majority of power is concentrated below the self-resonant frequency, so almost any capacitor with high voltage rating could be used. The exception is when you opt for a much higher switching frequency (MHz and higher) and faster rise time (~1 ns) to ensure very efficient power conversion. In that case, your PWM driver might excite a self-resonance, and low-ESL capacitors are needed.

For digital decoupling applications, where we need to ensure current drawn into a PCB's PDN is smooth, using low ESL capacitors helps ensure the PDN impedance is smooth out to higher frequency. The goal is to keep PDN impedance below some target value as a low impedance translates into a small voltage disturbance on the PDN. This is why outdated high speed design application notes will tell you to use three capacitors for decoupling each IC (10 nF, 1 nF, and 100 pF). For advanced components like [high speed FPGAs](#), which can have very low rise times, the decoupling strategy can be much more complex as we need flat impedance out to 10's or 100's of GHz.

What Determines Capacitor ESL and ESR?

There are three factors that contribute to the ESL and ESR values of a capacitor. These include:

- **Dielectric material:** The contact resistance between the dielectric and the capacitor lead determines the ESR value, and the permeability of the dielectric determines the ESR value.
- **Package size:** This factor has the greatest effect on ESL and ESR in a capacitor. Larger packages will have larger leads and contacts against the dielectric, so they can have larger ESL values.
- **Mounting style:** Through-hole components tend to have higher ESL than SMD capacitors due to the large size of the leads on through-hole capacitors.

Because the dielectric material used in the capacitor determines ESL and ESR, we can now see why some IC datasheets and application notes will recommend a specific type of capacitor. Certain types of capacitors (e.g., tantalum, ceramic, etc.) may tend to have lower self-resonant frequencies, so they are a better choice for use in high speed digital applications. Meanwhile, for power electronics, the use of larger capacitors is more about ensuring a high voltage rating and maintaining stable DC output, so ESL and self-resonance are less important.

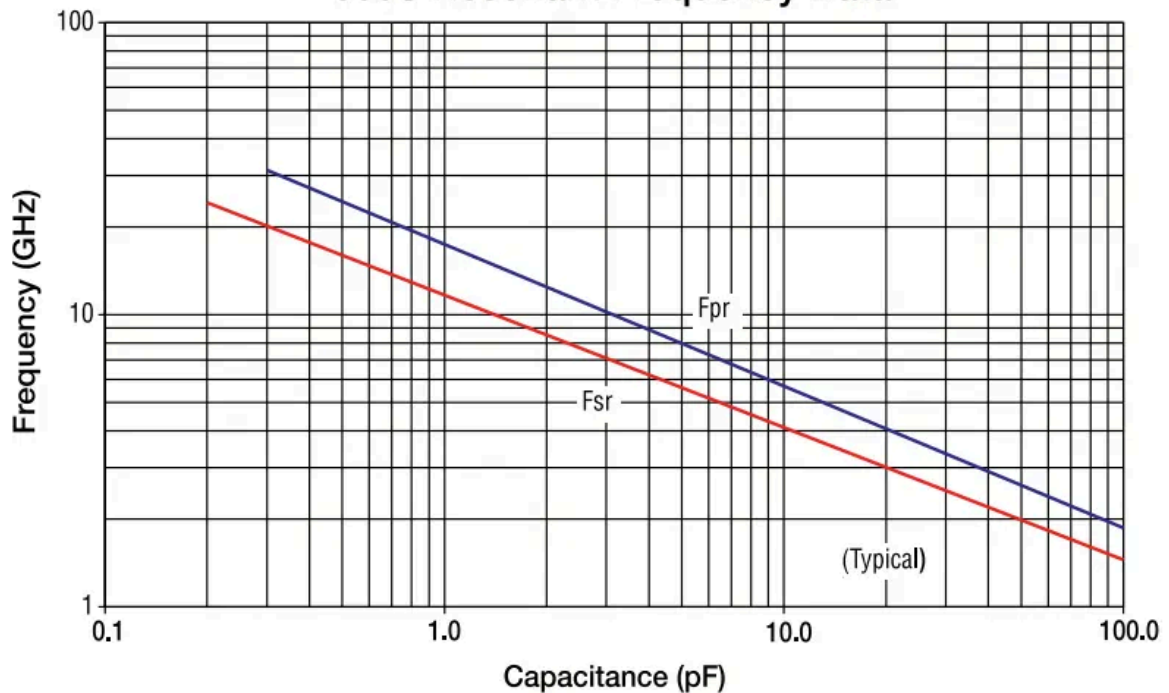
Selecting a Low-ESL Capacitor

Unfortunately, when you need to find a low-ESL capacitor, most datasheets do a poor job of giving you a specific value for ESL. Datasheets might do a better job showing an ESR value, which is important for understanding how flat the impedance curve is. Some datasheets for capacitors that are specifically marketed as high-frequency capacitors may include an impedance vs. frequency curve, which does help you immediately determine if the capacitor will meet your bandwidth requirements.

Identify a Candidate Low-ESL Capacitor

Because the ESL values of capacitors are rarely found in datasheets, you'll need to look at product guides from the manufacturer. If you can find a chart like that shown below, you can get a good idea of the ESL value for your capacitor. The following chart shows how self-resonance and capacitance are related for the [American Technical Ceramics 600 Series of MLCCs](#), and the slope of the curve is related to the ESL value of the capacitor.

600S Resonant Frequency Data



ATC 600 Series Data Sheet Test Condition Description

Capacitors horizontally mounted on 13.3-mil thick Rogers RO4350[®] softboard, 29-mil wide, 1/2 oz. Cu traces.
FSR = lowest frequency at which S11 response, referenced at capacitor edge, crosses real axis on Smith Chart.
FPR = lowest frequency at which there is a notch in S21 magnitude response.

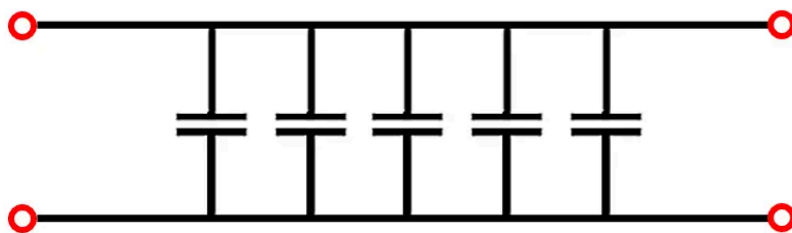
Digital vs. Analog

Selecting a low-ESL capacitor for an analog system, such as a [wireless system](#), is rather easy. Simply check that the capacitor acts like an ideal capacitor and that its self-resonant frequency is larger than the operating frequency in the system. Because digital signals are broadband, you need to compare the entire impedance vs. frequency curve to your signal bandwidth, you can't just look at a single frequency.

Decoupling with Low-ESL Capacitors

Remember, physically smaller capacitors have lower ESL values and thus higher self-resonant frequency; this is another reason why physically smaller capacitors are recommended for high-speed digital systems. If you look at the layout and PDN decoupling scheme in a typical high-speed digital system, you'll see there are multiple capacitors placed in parallel in the decoupling network. There is a specific reason for this: using multiples of the same capacitor in parallel will increase the total equivalent capacitance and decrease the PDN impedance, but it won't change the resonance frequency. This is shown in the example below for 5 capacitors with the same C and ESL values.

5 capacitors, each with same C and ESL



$$\text{Equivalent impedance: } Z_{eq} = \frac{1 - \omega^2 C \cdot ESL}{i5\omega C} \rightarrow \omega_r = \frac{1}{\sqrt{C \cdot ESL}}$$

I've ignored ESR in the above diagram, but we get the same result regardless; I'll leave this as an exercise for the reader. The point here is, if you need to select a low-ESL capacitor with high self-resonance frequency, you can use a smaller capacitance, and just put multiple capacitors in parallel. The frequency response for a single low-ESL capacitor or multiple identical capacitors in parallel will be the same.

The same ideas do not strictly apply to different capacitors with different C or ESL values placed in parallel. In this case, there will be multiple resonance peaks due to interaction between different RLC networks with different poles, and a more thorough analysis is needed to understand the impedance and frequency response of these capacitor networks.

Use the Best Search Engine to Find Low-ESL Capacitors

The best electronic parts search engine can help you narrow down to specific ESR values, mounting styles, dielectric materials, and of course capacitance value. When you can narrow it down to specific material types, you can identify different types of capacitors that are most likely to have low ESL values.

When you need to find low-ESL capacitors in standard packages from major manufacturers, use the comprehensive set of [advanced search and filtration features](#) in Octopart. When you use Octopart's electronics search engine, you'll have access to updated distributor pricing data, parts inventory, and parts specifications, and it's all freely accessible in a user-friendly interface. [Take a look at our passive components page](#) to find the components you need.

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